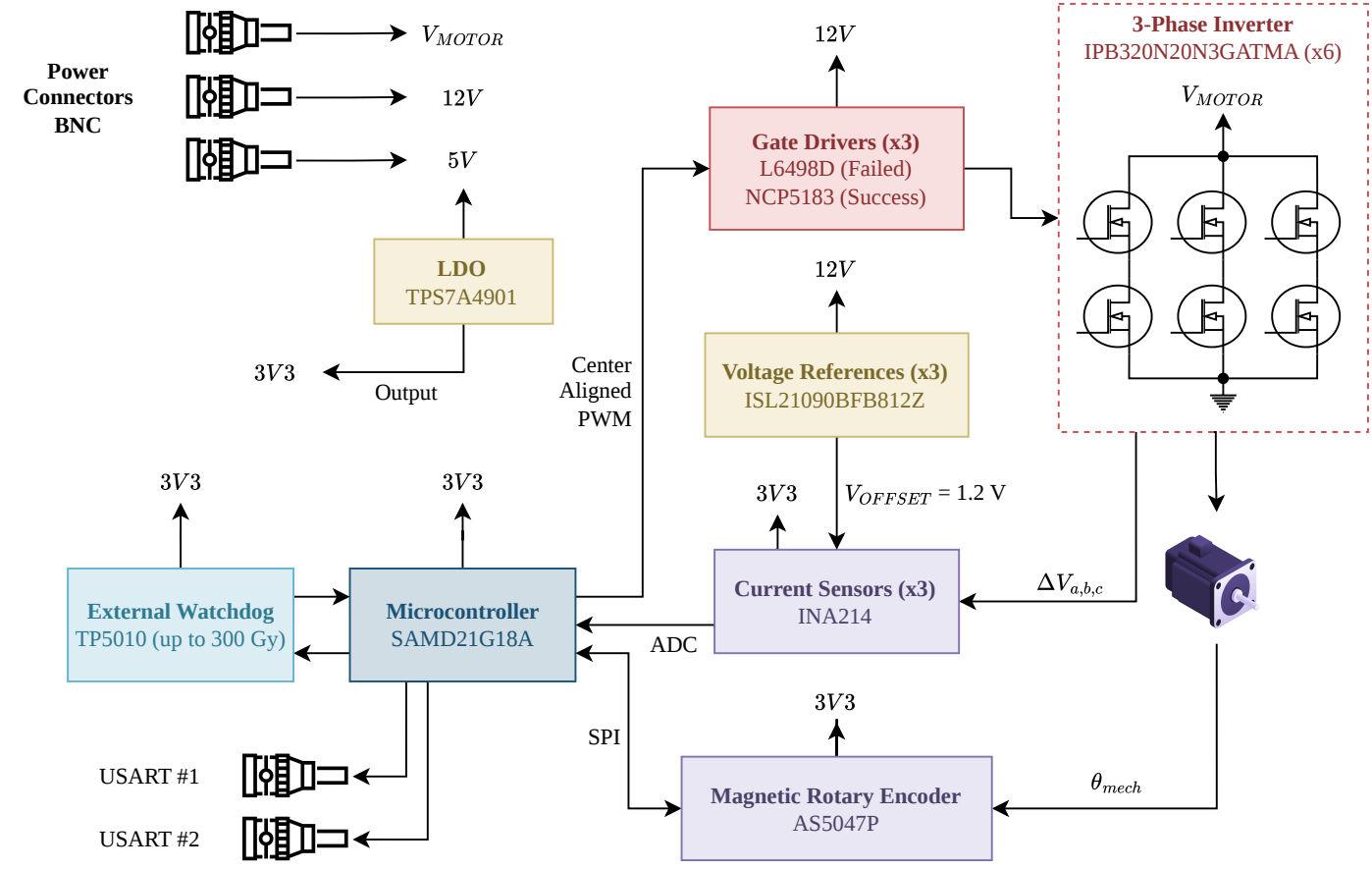

Power Connectors BNC
V_{MOTOR}
12V
5V
LDO
TPS7A4901
3V3
Output
12V
Gate Drivers (x3)
L6498D (Failed)
NCP5183 (Success)
3-Phase Inverter
IPB320N20N3GATMA (x6)
V_{MOTOR}
12V
Voltage References (x3)
ISL21090BFB812Z
Center Aligned PWM
3V3
V_{OFFSET} = 1.2 V
3V3
3V3
External Watchdog
TP5010 (up to 300 Gy)
Microcontroller
SAMD21G18A
Current Sensors (x3)
INA214
$\Delta V_{a,b,c}$
ADC
SPI
3V3
Magnetic Rotary Encoder
AS5047P
θ_{mech}
USART #1
USART #2